A First Course in Abstract Algebra

Section 44: Homological Algebra

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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An exact sequence has image equal to the next kernel at every term. We write 0 for the trivial additive group and use ordinary homology with integer coefficients. A relative group $H_k(X, Y)$ is computed from chains modulo the chains in Y; when Y is nonempty, comparison with X/Y uses reduced homology in dimension zero.

Exercise 44.1: An exact sequence with two nonzero terms

Exactness at A says $\ker f = \operatorname{im}(0 \to A) = 0$, so f is injective. Exactness at B says $\operatorname{im} f = \ker(B \to 0) = B$, so f is surjective. An injective and surjective homomorphism is an isomorphism. Thus the given map itself supplies $A \cong B$.

Exercise 44.2: A short exact sequence

(a) Exactness at C gives $\operatorname{im} j = \ker(C \to 0) = C$, so j is onto. **(b)** Exactness at A gives $\ker i = \operatorname{im}(0 \to A) = 0$, so i is injective and is an isomorphism from A to its image in B. **(c)** Exactness at B gives $\ker j = i[A]$. The first isomorphism theorem, together with surjectivity of j, gives $B/i[A] \cong C$, via $b + i[A] \mapsto j(b)$. Equality of the kernel with $i[A]$ proves that this coset formula is well defined and injective.

Exercise 44.3: Three equivalent conditions in an exact sequence

For the exact sequence $A \xrightarrow{i} B \xrightarrow{j} C \xrightarrow{k} D$, exactness says $\operatorname{im} i = \ker j$ and $\operatorname{im} j = \ker k$. Hence

$$\begin{aligned} i \text{ onto} &\iff \operatorname{im} i = B \iff \ker j = B \\ &\iff j = 0 \iff \operatorname{im} j = 0 \iff \ker k = 0 \iff k \text{ injective.} \end{aligned}$$

The middle condition $j = 0$ means it maps every element of B to the zero element of C. This proves all three requested conditions equivalent.

Exercise 44.4: When the middle map is an isomorphism

For $A \xrightarrow{g} B \xrightarrow{h} C \xrightarrow{i} D \xrightarrow{j} E \xrightarrow{k} F$, exactness yields

$$\operatorname{im} h = \ker i, \quad \operatorname{im} i = \ker j, \quad \operatorname{im} g = \ker h, \quad \operatorname{im} j = \ker k.$$

Thus $h = 0$ is equivalent to injectivity of i, and $j = 0$ is equivalent to surjectivity of i. Together these say exactly that i is an isomorphism $C \to D$, proving (a) $\iff$ (b). Also g is onto exactly when $\ker h = B$, namely when $h = 0$; and k is injective exactly when $\operatorname{im} j = 0$, namely when $j = 0$. This proves (a) $\iff$ (c), completing the equivalence.

Exercise 44.5: The induced-quotient-map principle

Exercise 14.39 gives the criterion: a homomorphism $f : G \to G'$ induces $G/H \to G'/H'$ by $gH \mapsto f(g)H'$ if $f(H) \subseteq H'$. For Theorem 44.4, the chain-map equation first shows $f_k(Z_k(A)) \subseteq Z_k(A')$, since

$\partial' f_k z = f_{k-1}\partial z = 0$. It also shows $f_k(B_k(A)) \subseteq B_k(A')$, since $f_k(\partial c) = \partial'(f_{k+1}c)$. Apply the quotient-map criterion to Z_k/B_k to obtain the induced homology map. For Theorem 44.7, $\partial_k(A'_k) \subseteq A'_{k-1}$ by the subcomplex assumption. Apply the same criterion with $G = A_k$, $H = A'_k$, $G' = A_{k-1}$, $H' = A'_{k-1}$, and $f = \partial_k$. This gives the well-defined quotient boundary $c + A'_k \mapsto \partial c + A'_{k-1}$. Its square is zero because $\partial^2 = 0$. Thus both theorems use precisely the same representative-independence mechanism as Exercise 14.39.

Exercise 44.6: Relative homology of a torus modulo a

After eliminating interior triangulation edges, use the torus chain model with one vertex, edges a, b, and one face F. Its boundaries are $\partial_1 = 0$ and $\partial_2 F = a + b - a - b = 0$. The circle subcomplex a contains the vertex and edge a. In the relative chain complex these are set to zero, leaving

$$C_2(X, a) = \mathbb{Z}F, \qquad C_1(X, a) = \mathbb{Z}b, \qquad C_0(X, a) = 0,$$

with zero boundary maps. Hence

$$H_2(X, a) \cong \mathbb{Z}, \quad H_1(X, a) \cong \mathbb{Z}, \quad H_0(X, a) = 0,$$

and higher groups vanish. Collapsing a gives a pinched torus with the positive-dimensional homology of $S^2 \vee S^1$. There is a necessary qualification to the printed parenthesis: relative homology corresponds to *reduced* homology of the quotient. The quotient's ordinary H_0 is $\mathbb{Z}$, whereas the relative H_0 computed here is zero.

Exercise 44.7: The exact sequence of the torus pair

Use $H_1(X) = \mathbb{Z}[a] \oplus \mathbb{Z}[b]$ and choose matching orientations for the whole-surface generators. The nonzero part of the exact sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{u} \mathbb{Z}^2$$
$$\xrightarrow{v} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \longrightarrow 0,$$

where the groups in order are $H_2(X)$, $H_2(X, a)$, $H_1(a)$, $H_1(X)$, $H_1(X, a)$, $H_0(a)$, $H_0(X)$. Here $u(t) = (t, 0)$ and $v(r, s) = s$. The first map carries the whole torus to the same relative 2-cycle. The first connecting map is zero because that 2-cycle has zero boundary. The second connecting map is zero because b already has zero boundary. To check exactness directly: the identity map has full image, equal to the next zero map's kernel; the zero image equals $\ker u = 0$; $\operatorname{im} u = \mathbb{Z} \times 0 = \ker v$; v is onto, matching the next zero map's full kernel; and the final identity has zero kernel. This checks every position.

Exercise 44.8: Relative homology of a Klein bottle modulo a

For the Klein-bottle polygon use one vertex, edges a, b, and a face F with $\partial_2 F = 2a$ and $\partial_1 = 0$. Set the vertex and the edge a to zero in the relative complex. Then $\partial_2 F$ also becomes zero, leaving one free generator in dimensions two and one and no dimension-zero generator. Thus

$$H_2(X, a) \cong \mathbb{Z}, \qquad H_1(X, a) \cong \mathbb{Z}, \qquad H_0(X, a) = 0,$$

with higher groups zero. This explains why collapsing the torsion loop creates a top-dimensional cycle: its original boundary was twice the collapsed edge. As in Exercise 44.6, the connected pinched quotient has ordinary $H_0 = \mathbb{Z}$; its *reduced* homology agrees with these relative groups.

Exercise 44.9: The exact sequence of the Klein-bottle pair

Here $H_2(X) = 0$, $H_1(X) = \mathbb{Z}_2[a] \oplus \mathbb{Z}[b]$, and $H_0(X) = \mathbb{Z}$. The relevant sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{u} \mathbb{Z}_2 \oplus \mathbb{Z}$$
$$\xrightarrow{v} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \longrightarrow 0.$$

The first $\mathbb{Z}$ is $H_2(X, a)$; the next is $H_1(a)$; the later three $\mathbb{Z}$ groups are $H_1(X, a)$, $H_0(a)$, $H_0(X)$. The connecting map sends F to its boundary $2a$, so it is multiplication by 2. Also $u(t) = (\overline{t}, 0)$ and $v(\overline{r}, s) = s$. Check exactness: multiplication by 2 is injective, $\ker u = 2\mathbb{Z}$, $\operatorname{im} u = \mathbb{Z}_2 \oplus 0 = \ker v$, v is onto, the following map is zero, and the final identity has zero kernel. These identities verify every term, including the torsion contribution.

Exercise 44.10: An annulus relative to both boundary circles

Let the boundary circles have the same angular orientation, and let Y be their union. A coherently oriented sum of the annulus's triangles has boundary equal to the difference of those two circles, which becomes zero modulo Y. It gives one relative 2-cycle F. A radial path e from one boundary component to the other has boundary in Y, so it gives one relative 1-cycle. Cutting the annulus along this path produces a disk; eliminating interior triangle edges shows that F and e exhaust the independent relative cycles. In a reduced relative chain model, there is one generator in degrees two and one, no generator in degree zero, and zero boundary maps. Hence

$$H_2(X, Y) \cong \mathbb{Z}, \qquad H_1(X, Y) \cong \mathbb{Z}, \qquad H_0(X, Y) = 0,$$

and higher groups vanish. The relative 1-generator is a radial path, not the circular loop, since that loop is homologous to a boundary circle and is zero in the relative group.

Exercise 44.11: The exact sequence for the annulus pair

Use the same angular orientation on the two boundary circles. Then inclusion maps $H_1(Y) = \mathbb{Z}^2 \to H_1(X) = \mathbb{Z}$ by $s(r, t) = r + t$. The analogous H_0 map is also s. Choose the orientation of the relative fundamental 2-chain and radial 1-chain so that their connecting maps are $d(k) = (k, -k)$. The nonzero part of the sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d} \mathbb{Z}^2 \xrightarrow{s} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{d} \mathbb{Z}^2 \xrightarrow{s} \mathbb{Z} \longrightarrow 0.$$

The groups are respectively $H_2(X, Y)$, $H_1(Y)$, $H_1(X)$, $H_1(X, Y)$, $H_0(Y)$, $H_0(X)$; $H_2(X)$ and $H_0(X, Y)$ are zero. Directly, d is injective, $\operatorname{im} d = \{(k, -k)\} = \ker s$, and s is onto. These facts check exactness at both pairs of $\mathbb{Z}^2$ and $\mathbb{Z}$ terms. At the middle relative $\mathbb{Z}$, the incoming zero map has image zero, equal to the next d's kernel; at the preceding $\mathbb{Z}$, the onto map s fills the kernel of the zero map.

Exercise 44.12: The quotient projection is a chain map

Let $j_k(c) = c + A'_k$. By the definition of the quotient boundary,

$$\overline{\partial}_k(j_k(c)) = \overline{\partial}_k(c + A'_k) = \partial_k c + A'_{k-1} = j_{k-1}(\partial_k c).$$

Thus $\overline{\partial}_k j_k = j_{k-1}\partial_k$ for every k. This proves the squares commute, so the quotient projections constitute a chain map.

Exercise 44.13: Well-definedness of the connecting map

A relative homology class is represented by $c \in A_k$ with $\partial c \in A'_{k-1}$. Since $\partial^2 c = 0$, this boundary is a cycle of A', so $[\partial c] \in H_{k-1}(A')$ is meaningful. Two choices representing the same relative homology class differ by

$$c' - c = \partial d + a', \qquad d \in A_{k+1}, \quad a' \in A'_k.$$

Indeed equality modulo relative boundaries is precisely this condition, including the possible change of coset representative in A_k/A'_k. Taking boundaries gives $\partial c' - \partial c = \partial a'$, a boundary *inside* A'. Thus $[\partial c'] = [\partial c]$ there, proving independence of both choices. Finally $\partial(c_1 + c_2) = \partial c_1 + \partial c_2$, so the rule $[c] \mapsto [\partial c]$ preserves addition. It is therefore a well-defined homomorphism.

Exercise 44.14: Consecutive homology-sequence maps compose to zero

There are three kinds of consecutive composites to check. First $j_* i_* [z'] = [z' + A'_k] = 0$, since a chain in the subcomplex is zero in the quotient. Second $\partial_* j_* [z] = [\partial z] = 0$, because a homology representative z in A is a cycle. Third $i_* \partial_* [c] = [\partial c] = 0$ in $H_{k-1}(A)$, because ∂c is a boundary in A (even though it need not be a boundary in A'). Thus every consecutive composite is zero. All groups are abelian and all maps are homomorphisms, so the long sequence is indeed a chain complex.

Exercise 44.15: Exactness of the long homology sequence

Use $[z]$ for an absolute homology class, and use $[c]_{\text{rel}}$ for a relative class, with $\partial c \in A'$.

(a) If a class comes from $H_k(A')$, its chain is zero in A_k/A'_k. Thus $\operatorname{im} i_* \subseteq \ker j_*$.

(b) If $j_*[z] = 0$, the cycle $z \in A_k$ satisfies $z = \partial d + a'$ with $d \in A_{k+1}$ and $a' \in A'_k$. Taking boundaries gives $\partial a' = 0$. Hence $[z] = [a'] = i_*[a']$ in $H_k(A)$, so $\ker j_* \subseteq \operatorname{im} i_*$.

(c) For an absolute cycle z, $\partial_* j_* [z] = [\partial z] = 0$. Thus $\operatorname{im} j_* \subseteq \ker \partial_*$.

(d) If $\partial_* [c]_{\text{rel}} = 0$, there is $a' \in A'_k$ with $\partial c = \partial a'$. Then $z = c - a'$ is a cycle in A, and $j_*[z] = [c]_{\text{rel}}$ since a' vanishes in the quotient. Therefore $\ker \partial_* \subseteq \operatorname{im} j_*$.

(e) The class $\partial_* [c]_{\text{rel}} = [\partial c]$ maps to zero in $H_{k-1}(A)$ because it bounds c there. Thus $\operatorname{im} \partial_* \subseteq \ker i_*$.

(f) If $[a'] \in H_{k-1}(A')$ maps to zero in $H_{k-1}(A)$, then $a' = \partial c$ for some $c \in A_k$. Its boundary is in A', so it represents a relative cycle, and $\partial_* [c]_{\text{rel}} = [a']$. Thus $\ker i_* \subseteq \operatorname{im} \partial_*$.

The six inclusions pair into equality at all three types of term, in every dimension. This proves exactness of the entire sequence.

Exercise 44.16: Chain homotopies induce equal homology maps

Let $z \in Z_k(A)$ represent a homology class. Because $\partial z = 0$, the given homotopy identity reduces to

$$f_k(z) - g_k(z) = \partial'_{k+1}(D_k z) + D_{k-1}(0) = \partial'_{k+1}(D_k z).$$

Thus the two image cycles differ by a boundary in A', so they represent the same class in $H_k(A')$. Since every homology class has a cycle representative, $f_{*k} = g_{*k}$ for every k. The chain-map hypotheses

ensure both induced maps are defined; the homotopy supplies the boundary witnessing equality on each class.